

Report 6

Monte Carlo Simulation in Finance

Black-Scholes Models, Path-Dependent & Exotic Options, Worst-Of Certificate

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1. Model Setup and Parameters

All simulations in this report are based on the Black-Scholes (BS) market model for an underlying asset following a Geometric Brownian Motion (GBM). The baseline parameters used throughout the exercises are:

Parameter	Symbol	Value
Initial asset price	S_0	100
Drift / Risk-free rate	r (or μ)	1% (simulations: 5%)
Volatility	σ	20%
Time to maturity	T	1 year
Strike price	K	99
Time step	dt	1/252 (daily)
Discretisation steps	N_{steps}	252

Under the risk-neutral measure, the GBM dynamics are given by the stochastic differential equation:

$$dS_t = r \cdot S_t \cdot dt + \sigma \cdot S_t \cdot dW_t$$

where W_t is a standard Brownian motion. The closed-form solution is:

$$S_T = S_0 \cdot \exp[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} \cdot Z], \quad Z \sim N(0, 1)$$

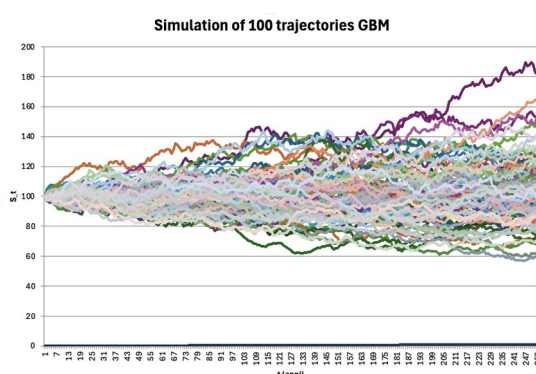
This analytical solution is used both as a direct pricing tool (single-step simulation) and as the benchmark for discretised multi-step approaches.

2. GBM Trajectory Simulation

The first exercise implements a VBA macro that simulates $N = 100$ trajectories of the GBM over one year with 252 daily time steps. Each path starts at $S_0 = 100$ and evolves via the Euler-Maruyama discretisation:

$$S_{t+dt} = S_t + r \cdot S_t \cdot dt + \sigma \cdot S_t \cdot \sqrt{dt} \cdot Z_i, \quad Z_i \sim N(0, 1)$$

The resulting chart visualises the dispersion of paths over time.



These simulated trajectories form the foundation for all subsequent path-dependent option pricing in Exercises 6–8.

3. Vanilla Option Pricing via Monte Carlo

3.1 Theory: Risk-Neutral Pricing

Under the risk-neutral measure, the price of a European call option with strike K and maturity T is:

$$C = e^{-rT} \cdot E^Q[\max(S_T - K, 0)]$$

Monte Carlo approximates this expectation as a sample average over N simulated terminal values $S_T^{(i)}$:

$$\hat{C} = e^{-rT} \cdot (1/N) \sum_i \max(S_T^{(i)} - K, 0)$$

The analytical benchmark is given by the Black-Scholes formula:

$$C_{BS} = S_0 \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$$

$$d_1 = [\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T] / (\sigma\sqrt{T}), \quad d_2 = d_1 - \sigma\sqrt{T}$$

For the given parameters ($S_0=100$, $K=99$, $r=1\%$, $\sigma=20\%$, $T=1$), the Black-Scholes price yields:

$$C_{BS} \approx 8.92, \quad P_{BS} \approx 6.93$$

3.2 Three Simulation Schemes

Three approaches were implemented and compared:

Exercise 2 — Single-Step (Closed Form of S_T)

N terminal values are drawn directly from the log-normal distribution using the exact closed-form solution. This is the most accurate MC approach as it uses no discretisation error:

$$S_T = S_0 \cdot \exp[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} \cdot N(0, 1)]$$

Exercise 3 — Multi-Step Euler Scheme

The GBM SDE is discretised into 252 steps using the Euler-Maruyama scheme. Each step introduces a discretisation error of order $O(dt)$, making it less precise than the exact form for the same N :

$$S_{t+dt} = S_t \cdot (1 + r \cdot dt + \sigma \cdot \sqrt{dt} \cdot Z)$$

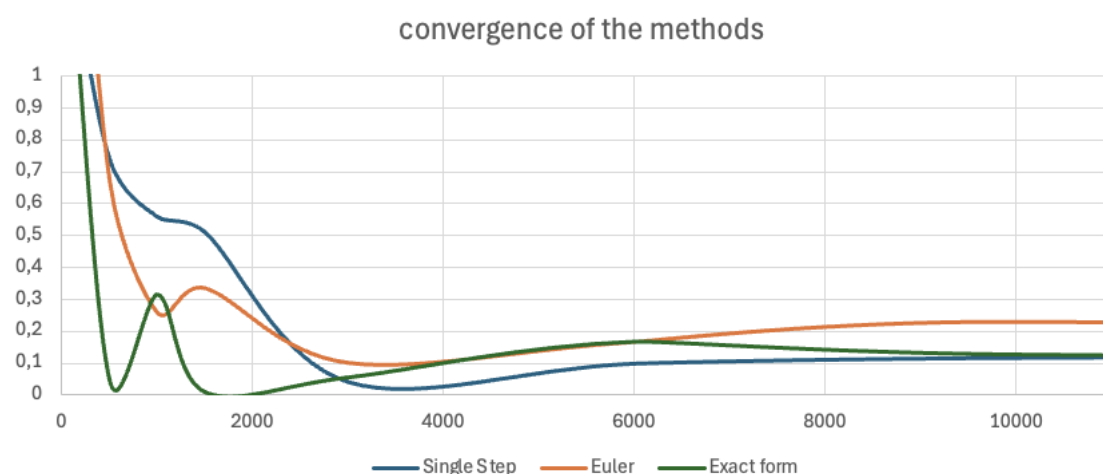
Exercise 4 — Multi-Step Exact Form

Instead of the Euler scheme, the exact multiplicative update is used at each step, eliminating discretisation error while retaining the path-dependent structure:

$$S_{t+dt} = S_t \cdot \exp[(r - \frac{1}{2}\sigma^2)dt + \sigma \cdot \sqrt{dt} \cdot Z]$$

3.3 Convergence Analysis and Results

The convergence table below, taken directly from the spreadsheet simulation outputs, compares the three methods against the Black-Scholes benchmark (≈ 8.92) across increasing sample sizes:



Key findings from the convergence analysis:

- All three methods converge toward the Black-Scholes price as N grows. The law of large numbers guarantees convergence, while the central limit theorem implies a standard error of order $\sigma_{MC} / \sqrt{N}$.
- The Exact Form (consistently achieves lower absolute errors at high N, confirming that eliminating discretisation error matters, especially at moderate sample sizes).
- The Euler Scheme (shows higher variance at low N due to accumulated discretisation errors across 252 steps, but becomes competitive above $N \approx 3,000$).
- Single-Step has the best theoretical accuracy per simulation since it draws directly from the correct marginal distribution, but **all methods are statistically equivalent at $N \geq 10,000$** .
- We notice that for European vanilla options, the single-step and multi-step exact simulations are mathematically equivalent and consistently optimal while the Euler-Maruyama scheme introduces a discretisation bias that strictly requires a higher number of time steps to converge to the true price.

4. Asian Options

4.1 Theory

Asian options are path-dependent derivatives whose payoff depends on the arithmetic average of the asset price over the option's life, rather than the terminal value alone. The arithmetic average Asian call payoff is:

$$\text{Payoff} = \max\left(\left(\frac{1}{T}\right) \int_0^T S_t dt - K, 0\right) \approx \max\left(\left(\frac{1}{N}\right) \sum S_{\{t_i\}} - K, 0\right)$$

Asian options have two important properties that distinguish them from vanilla options:

- Averaging reduces the effective volatility: the arithmetic average of a GBM process has lower variance than the terminal value, since averaging smooths out fluctuations. This makes Asian options systematically cheaper than their vanilla equivalents.
- No closed-form formula exists for arithmetic Asian options under BS (the sum of log-normals is not log-normal). The geometric average Asian option does have an analytical solution (Kemna-Vorst formula), which serves as a control variate or benchmark. (

4.2 Benchmark: Kemna-Vorst (Geometric Average)

The Kemna-Vorst (1990) formula prices the geometric average Asian option analytically. This serves as a lower bound and validation check, since the geometric average is always $\leq$ the arithmetic average by Jensen's inequality. Reference: <https://docs.dises.univpm.it/web/quaderni/pdf/085.pdf> (Chapter 3)

$$C_{geo} \text{ (Kemna-Vorst)} \approx 5.140 \quad P_{geo} \approx 3.985$$

4.3 Results

The Monte Carlo arithmetic Asian option pricer (N = 5,000 simulations, 252 steps) yields:

Method	Call Price	Put Price
MC Arithmetic (this run)	5.389	3.912
Kemna-Vorst Geometric (benchmark)	5.140	3.985
B&S Vanilla (reference)	8.92	6.93

Stability across repeated runs (N = 5,000 each):

Run	Call	Put
1	5.383	3.794
2	5.514	3.853
3	5.481	3.772
4	5.254	3.823
5	5.279	3.764
10	5.394	3.809

The results confirm the theoretical expectation: arithmetic Asian calls (≈ 5.39) are cheaper than vanilla calls (≈ 8.92), as averaging reduces effective volatility. The arithmetic MC price slightly exceeds the geometric benchmark (5.14), consistent with Jensen's inequality. The standard deviation across runs is approximately ± 0.10 , indicating good stability at N = 5,000.

5. Other Path-Dependent Options

5.1 Lookback Options

A lookback option's payoff depends on the realised minimum or maximum of the asset price over the option's life:

$$\text{Call (floating): Payoff} = S_T - \min(S_t, t \leq T)$$

$$\text{Put (floating): Payoff} = \max(S_t, t \leq T) - S_T$$

Lookback options are more expensive than vanillas because they always capture the most favourable price path realisation. An analytical benchmark is available for continuous monitoring (Goldman-Sosin-Gatto formula, <https://docs.dises.univpm.it/web/quaderni/pdf/085.pdf> (chapetr 3)):

Method	Call Price	Put Price
MC (N=5,000, 252 steps)	14.804	15.589
Analytical Benchmark	15.414	16.409

The MC prices are slightly below the analytical benchmark, which is expected: discrete monitoring underestimates the true continuous lookback price. The discretisation bias is approximately $(1 - 0.5826 \cdot \sigma \cdot \sqrt{dt})$ per step (Convergence across 5 independent runs confirms the MC is stable with standard deviation $\approx \pm 0.10$).

5.2 Barrier Options

Barrier options are activated (knock-in) or deactivated (knock-out) when the asset price crosses a pre-specified barrier level B. Two types were implemented:

Down-and-Out Call (B = 90)

The option expires worthless if the asset price ever falls below B = 90. This reduces the vanilla call price since some scenarios that would have been profitable are knocked out.

$$\text{Payoff} = \max(S_T - K, 0) \cdot 1_{\{\min(S_t) > B\}}$$

Method	Price
MC (N=5,000)	7.245
Analytical Benchmark	7.215

Up-and-Out Put (B = 110)

The put expires worthless if the asset price rises above $B = 110$. This reduces the put price since high-price paths (where the put would be less valuable anyway) are knocked out.

$$\text{Payoff} = \max(K - S_T, 0) \cdot 1_{\{\max(S_t) < B\}}$$

Method	Price
MC (N=5,000)	5.411
Analytical Benchmark	5.377

Both barrier option prices match their analytical benchmarks with high accuracy (errors < 1%), validating the MC implementation. The discrete barrier monitoring introduces a small upward bias relative to continuous monitoring, visible in both cases. (references for closed formula: <https://docs.dises.univpm.it/web/quaderni/pdf/085.pdf> (Chapter 3))

6. Worst-Of Certificate — Multi-Asset Monte Carlo (Exercise 8)

6.1 Asset Selection and Historical Calibration

Three French large-cap equities listed on Euronext Paris were selected from Refinitiv (6 months of daily data, approx. 126 trading days):

- LVMH (LVMH.PA) — luxury goods conglomerate
- Pernod Ricard (PRTP.PA) — spirits and wines
- Hermès International (HRMS.PA) — luxury goods

Historical volatilities and correlations were estimated from daily log-returns:

Asset	Daily StdDev	Annualised Volatility
LVMH	1.742%	27.66%
Pernod Ricard	2.651%	42.08%
Hermès	1.958%	31.08%

Correlation matrix (Pearson, from daily log-returns):

	LVMH	PRTP	HRMS
LVMH	1.000	0.546	0.648
PRTP	0.546	1.000	0.634
HRMS	0.648	0.634	1.000

The correlations are significant ($\rho \approx 0.55$ – 0.65), reflecting the common luxury-goods sector exposure. Pernod Ricard shows the highest volatility (42%), driven by macro headwinds facing the spirits sector in 2024.

6.2 Correlated GBM Simulation — Cholesky Decomposition

To simulate correlated asset paths, the Cholesky decomposition of the correlation matrix Σ is computed: $\Sigma = L \cdot L^T$. Three independent standard normal variates $Z = (Z_1, Z_2, Z_3)$ are transformed into correlated variates: $\hat{Z} = L \cdot Z$. Each asset then evolves as:

$$S_i(t+dt) = S_i(t) \cdot \exp[(r - \frac{1}{2}\sigma_i^2)dt + \sigma_i \cdot \sqrt{dt} \cdot \hat{Z}_i]$$

The risk-free rate used is the 12-month Euribor: $r = 2.72\%$ (as of the data collection date).

6.3 Worst-Of Certificate Structure

A Worst-Of Cash Collect certificate is a structured product whose return depends on the worst-performing underlying over the observation period. The structure implemented follows this example and certificate: <https://www.onemarkets.it/it/onemarkets/formazione/cash-collect-worst-of/cash-collect-worst-of-esempio.html>

- Nominal: 100
- Maturity: $T = 3$ years
- Barrier: 70% of initial price
- Coupon: 23% (reverse-engineered to match market price ≈ 96)
- Autocall: triggered at $T = 1$ year if all underlyings are above initial level
- Autocall frequency: annual
- $N = 10,000$ simulations

Payoff logic at each observation date:

- Autocall: if $\min(S_1(t)/S_1(0), S_2(t)/S_2(0), S_3(t)/S_3(0)) \geq 1$, the product is called and returns Nominal + Coupon. (we notice that it is mandatory for the contract to be exercised, unlike american options, which makes the payoff simpler)
- At maturity: if the worst performer is above the barrier (70%), return Nominal + Coupon. If the worst performer is below the barrier, the investor receives the performance of the worst asset (capital at risk).

6.4 Pricing Results

The European Worst-Of price ($N = 10,000$, $T = 3$ years) with the calibrated parameters:

Run	Price
1	96.41
2	96.58
3	96.49
4	97.01
5	96.40
6	97.35
7	96.16
8	96.58
9	96.69
10	96.06
Mean (approx.)	96.48

The Monte Carlo price converges to approximately 96.48 (out of nominal 100), consistent with typical market pricing of Worst-Of certificates slightly below par. This validates the calibration choice (coupon = 23%), which was selected by reverse engineering to match the observed market price of around 96.

The high coupon (23%) relative to standard products reflects the elevated volatility of PRTP (42%), which significantly increases the probability that the worst performer breaches the barrier, compensating the investor for bearing this risk.

From a risk perspective, the correlation structure plays a key role: the relatively high pairwise correlations (0.55–0.65) reduce the probability of a large divergence between assets, which would make the worst-of

outcome more severe. Lower correlations would increase the product's riskiness and require a higher coupon to maintain the same price.

7. Conclusions

This report demonstrates the implementation and analysis of Monte Carlo simulation techniques applied to derivative pricing within the Black-Scholes framework. The key findings are:

- GBM simulation: 100 trajectories correctly capture the log-normal distribution of asset prices, with path dispersion growing as $\sigma\sqrt{t}$.
- Vanilla options: All three MC schemes converge to the B&S price. The exact-form multi-step method shows the lowest discretisation error. At $N \geq 10,000$, all methods are statistically equivalent.
- Asian options: MC correctly prices arithmetic Asians at approximately 5.39 (call), below the vanilla price of 8.92, validating the volatility-smoothing effect of averaging.
- Lookback and barrier options: Results closely match analytical benchmarks, with small biases attributable to discrete monitoring.
- Worst-Of certificate: Multi-asset correlated simulation via Cholesky decomposition yields a stable price of ≈ 96.48 , consistent with market conventions. High coupon driven by PRTP's elevated volatility.

A recurring theme is the trade-off between computational cost (N) and precision: MC standard error scales as $1/\sqrt{N}$, meaning a $10\times$ improvement in accuracy requires $100\times$ more simulations. Variance reduction techniques (antithetic variates, control variates) would be natural extensions to improve efficiency.

ADVANTAGES OF MONTE CARLO

Monte Carlo simulation easily overcomes the curse of dimensionality inherent in multi-asset derivatives, whereas closed-form formulas fail to efficiently solve high-dimensional integrals. Furthermore, it seamlessly evaluates complex path-dependent features—such as discrete monitoring, memory effects, and autocall triggers—that disrupt the continuity required by analytical models. Finally, this numerical framework provides the ultimate flexibility to incorporate advanced market dynamics, like stochastic volatility or jump-diffusion, without requiring a complete mathematical reformulation.

REFERENCES for the scripts

- **Black Scholes, Monte Carlo (1 step):** VinegarHill FinanceLab
- **Multi steps Monte Carlo:** ("A Modern Approach to Teaching Options Pricing Using Monte Carlo Simulation with Python" Wesley T. Tzeng, 2025) – I then adapted the script to VBA
- **Other options:** Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. - **Hull, J. C. (2018).** *Options, Futures, and Other Derivatives*. I adapted the multisteps code for the different payoffs, also with AI help, but always checking the validity of the script against a benchmark defined by a closed formula found in <https://docs.dises.univpm.it/web/quaderni/pdf/085.pdf> (chapter 3).